Maximum and minimum values

1

a $x = 1, 3$ **b** $(1, 11)$ **c** $(3, 3)$ **d** 227

e 3

2

a $(0, 4)$

b $\left(-\frac{3}{5}, \frac{91}{25}\right), \left(-\frac{1}{3}, \frac{97}{27}\right)$

c $\left(-\frac{3}{5}, \frac{91}{25}\right)$ is a local maximum and $\left(-\frac{1}{3}, \frac{97}{27}\right)$ is a local minimum.

d As $f''(x) = 0$ and the concavity changes from negative to positive, there is a point of inflection at $x = -\frac{7}{15}$. The point of inflection is $\left(-\frac{7}{15}, \frac{2441}{675}\right)$.

e As $x \to \infty, y \to \infty$ and as $x \to -\infty, y \to -\infty$.

f The minimum is 3 and the maximum is 19.

3

a $(0, 5)$

b None

c Even

d $(-1, 4), (0, 5), (1, 4)$

e $(0, 5)$ is a local maximum, and $(1, 4)$ and $(-1, 4)$ are global minimums.

f As $f''(x) = 0$ and the concavity changes from positive to negative at $x = -\frac{\sqrt{3}}{3}$ and from negative to positive at $x = \frac{\sqrt{3}}{3}$. The points of inflection are $\left(-\frac{\sqrt{3}}{3}, \frac{40}{9}\right)$ and $\left(\frac{\sqrt{3}}{3}, \frac{40}{9}\right)$.

g As $x \to \infty, y \to \infty$ and as $x \to -\infty, y \to \infty$.

h The maximum is 13 and the minimum is 4.

4

a $(0, -1)$

b $(-2, -33), (0, -1), (1, -6)$

c $(0, -1)$ is a local maximum, $(1, -6)$ is a local minimum and $(-2, -33)$ is a global minimum.

d As $f''(x) = 0$ and the concavity changes from positive to negative at $x = -1.22$ and from negative to positive at $x = 0.55$. The points of inflection are $(-1.22, -19.36)$ and $(0.55, -3.68)$.

e As $x \to \infty, y \to \infty$ and as $x \to -\infty, y \to \infty$.

f The maximum is -1 and the minimum is -14.

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Graphs of functions

5

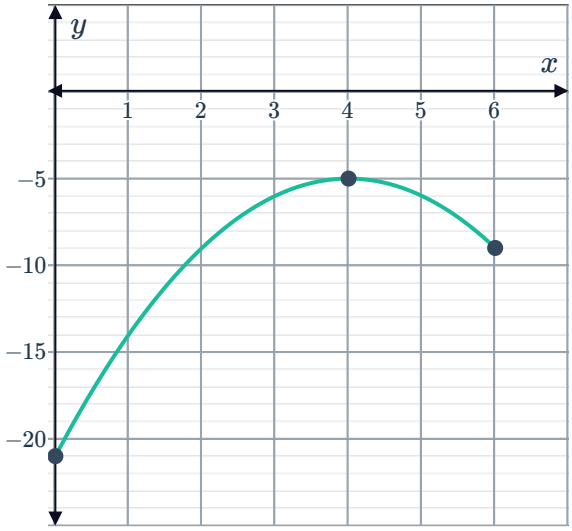
a $(0, -21)$

b $(4, -5)$

c $(4, -5)$ is a global maximum point.

d The maximum value is -5 and the minimum value is -21.

e



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6

a (0,1)

b $(-1,0),(1,0)$

c Even

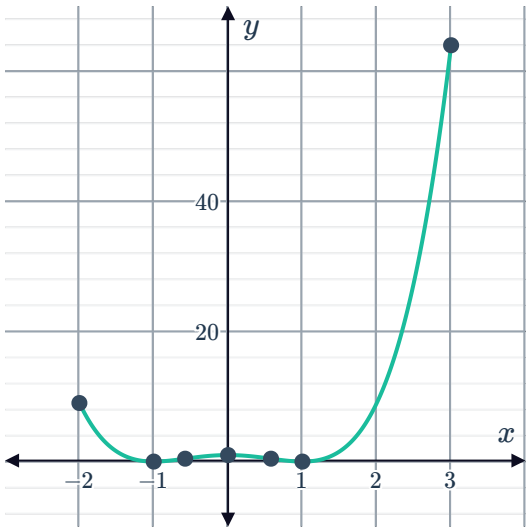
d $(-1,0),(0,1),(1,0)$

e $(-1,0)$ and $(1,0)$ are global minimum and $(0,1)$ is a local maximum.

f As $f''(x)=0$ and the concavity changes from positive to negative at $x=-\frac{\sqrt{3}}{3}$ and negative to positive at $x=\frac{\sqrt{3}}{3}$. The points of inflection are $\left(-\frac{\sqrt{3}}{3},\frac{4}{9}\right)$ and $\left(\frac{\sqrt{3}}{3},\frac{4}{9}\right)$.

g The maximum is 64 and the minimum is 0.

h



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7

a $(0, 1)$

b $\left(\frac{1}{2}, 0\right), (1, 0)$

c $\left(\frac{1}{2}, 0\right), \left(\frac{5}{6}, \frac{2}{27}\right)$

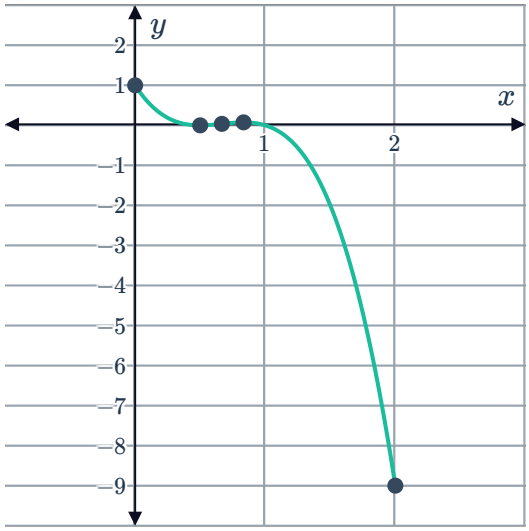
d $\left(\frac{1}{2}, 0\right)$ is a local minimum and $\left(\frac{5}{6}, \frac{2}{27}\right)$ is a local maximum.

e As $f''(x) = 0$ and the concavity changes from positive to negative, there is a point of inflection at $x = \frac{2}{3}$. The point of inflection is $\left(\frac{2}{3}, \frac{1}{27}\right)$.

f As $x \to \infty, y \to -\infty$ and as $x \to -\infty, y \to \infty$.

g The maximum is 1 and the minimum is -9.

h



8

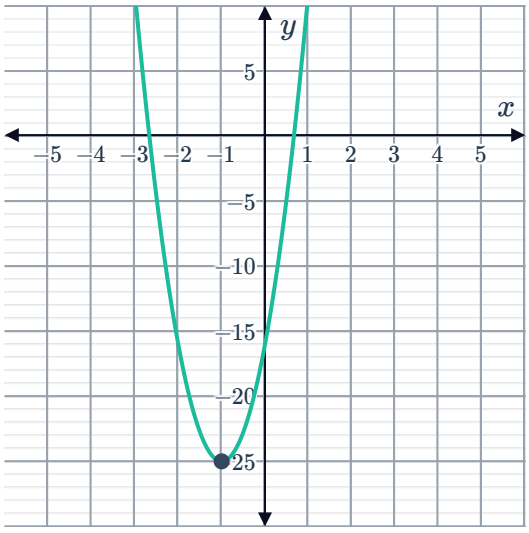
a $(0, -16)$

b $\left(-\frac{8}{3}, 0\right), \left(\frac{2}{3}, 0\right)$

c $(-1, -25)$

d $(-1, -25)$ is the global minimum point.

e



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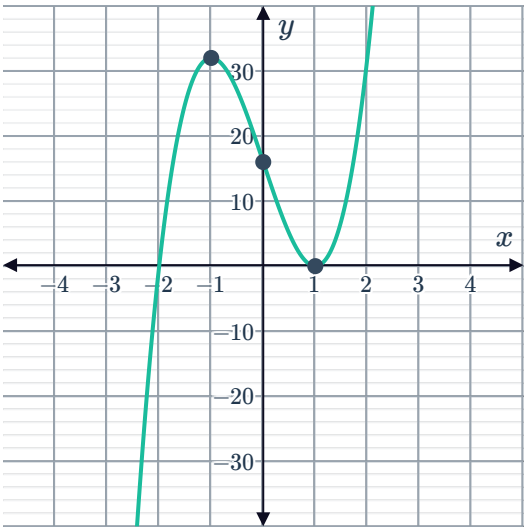
9

a $(-1, 32), (1, 0)$

b $(-1, 32)$ is a local maximum and $(1, 0)$ is a local minimum.

c $(0, 16)$

d



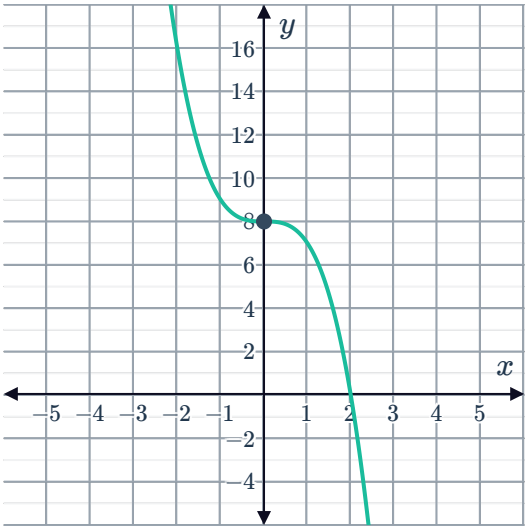
10

a

i $(0, 8)$ is a horizontal point of inflection.

ii $(0, 8)$

iii



iv $(-\infty, \infty)$

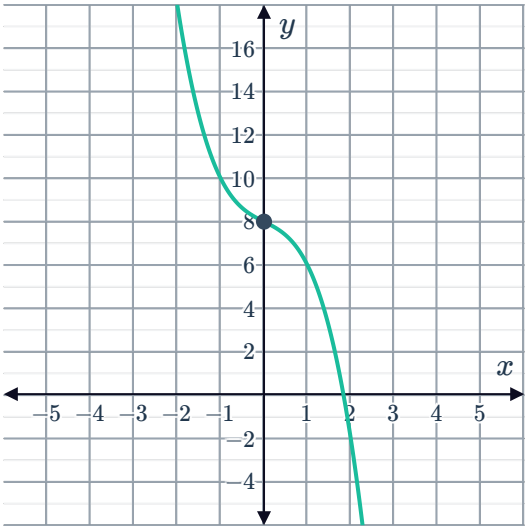
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b

i None

ii $(0, 8)$

iii



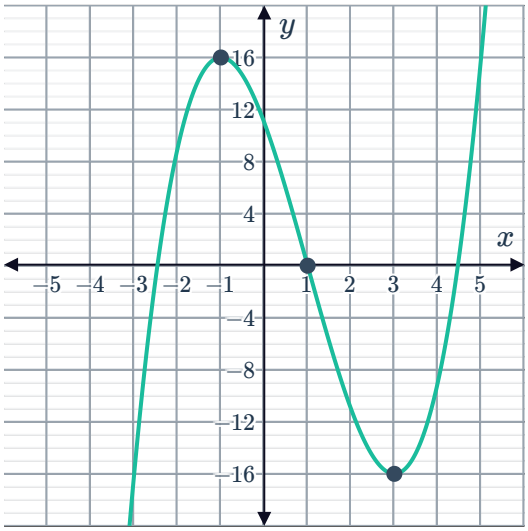
iv $(-\infty, \infty)$

c

i $(-1, 16)$ is a local maximum and $(3, -16)$ is a local minimum.

ii $(1, 0)$

iii



iv $(-1, 3)$

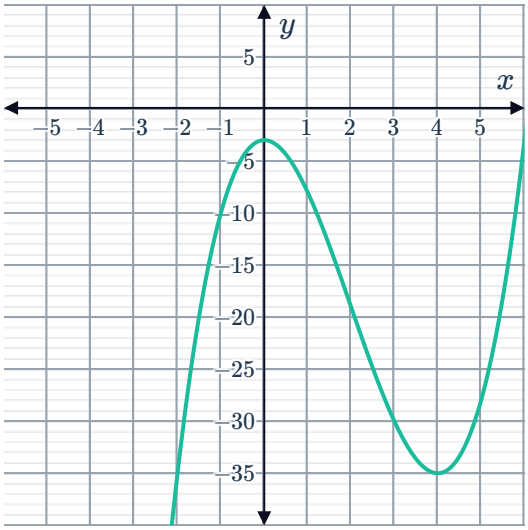
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d

i $(0, -3)$ is a local maximum and $(4, -35)$ is a local minimum.

ii $(2, -19)$

iii



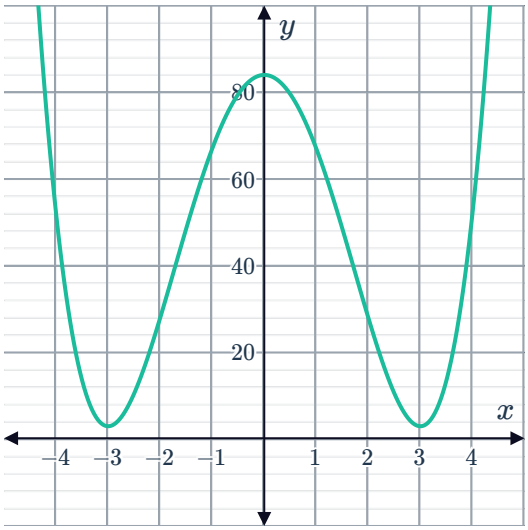
iv $(0, 4)$

e

i $(-3, 3)$ and $(3, 3)$ are global minimums, and $(0, 84)$ is a local maximum

ii $\left(-\sqrt{3}, 39\right), \left(\sqrt{3}, 39\right)$

iii



iv $(-\infty, -3) \cup (0, 3)$

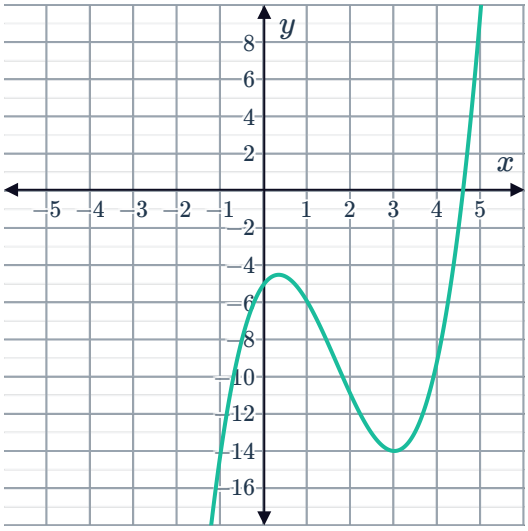
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f

i $\left(\frac{1}{3}, -\frac{122}{27}\right)$ is a local maximum and $(3, -14)$ is a local minimum.

ii $\left(\frac{5}{3}, -\frac{250}{27}\right)$

iii



iv $\left(\frac{1}{3}, 3\right)$

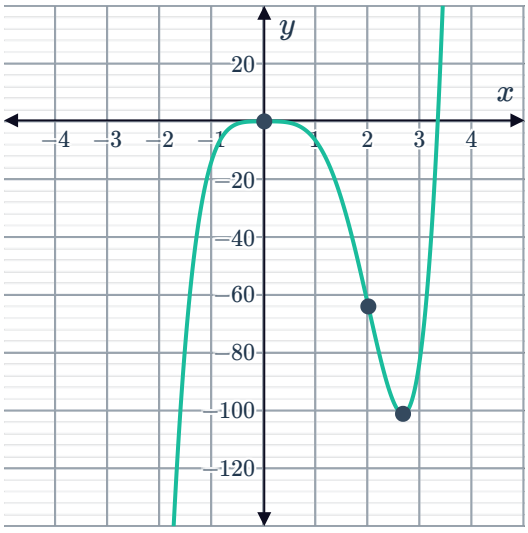
11

a

i $(0, 0)$ is a local maximum and $\left(\frac{8}{3}, -\frac{8192}{81}\right)$ is a local minimum.

ii $(2, -64)$

iii



iv $(-\infty, 0) \cup \left(\frac{8}{3}, \infty\right)$

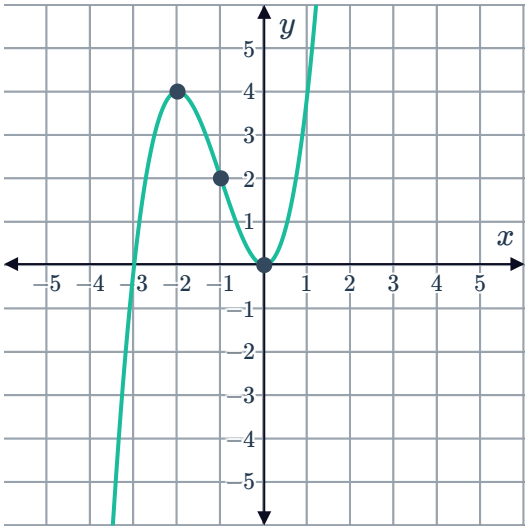
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b

i $(-2, 4)$ is a local maximum and $(0, 0)$ is a local minimum.

ii $(-1, 2)$

iii



iv $(-\infty, -2) \cup (0, \infty)$

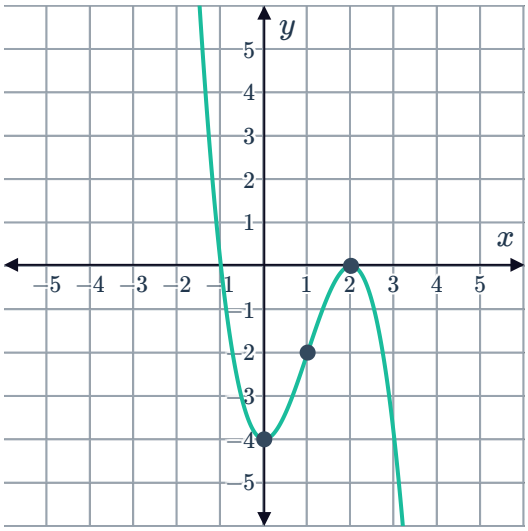
12

a

i $(2, 0)$ is a local maximum and $(0, -4)$ is a local minimum.

ii $(1, -2)$

iii



iv Concave up on $(-\infty, 1)$ since $f''(x)$ is positive and concave down on $(1, \infty)$ since $f''(x)$ is negative

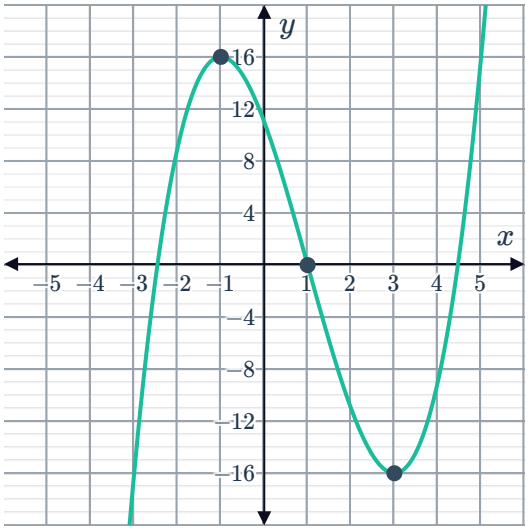
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b

i $(-1, 16)$ is a local maximum and $(3, -16)$ is a local minimum.

ii $(1, 0)$

iii



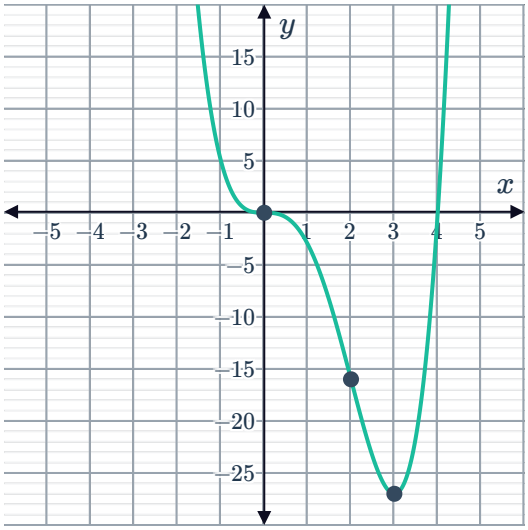
iv Concave up on $(1, \infty)$ since $f''(x)$ is positive and concave down on $(-\infty, 1)$ since $f''(x)$ is negative

c

i $(0, 0)$ is a horizontal point of inflection and $(3, -27)$ is the global minimum.

ii $(0, 0), (2, -16)$

iii



iv Concave up on $(-\infty, 0)$ and $(2, \infty)$ since $f''(x)$ is positive and concave down on $(0, 2)$ since $f''(x)$ is negative

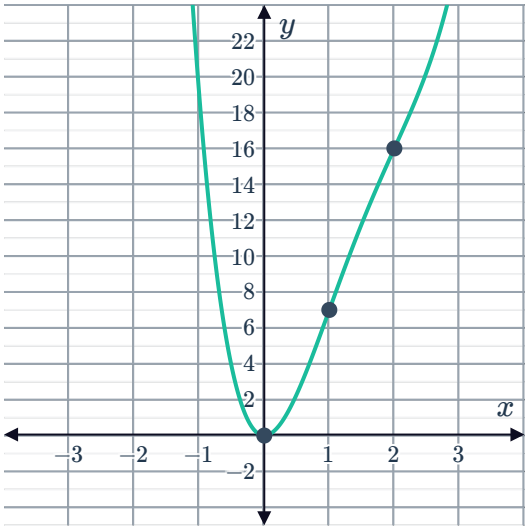
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d

i $(0, 0)$ is the global minimum.

ii $(1, 7), (2, 16)$

iii



iv Concave up on $(-\infty, 1)$ and $(2, \infty)$ since $f''(x)$ is positive and concave down on $(1, 2)$ since $f''(x)$ is negative

13

a $(0, 253)$

b $\left(-\sqrt[3]{\frac{3}{2}} - 5, 0\right)$

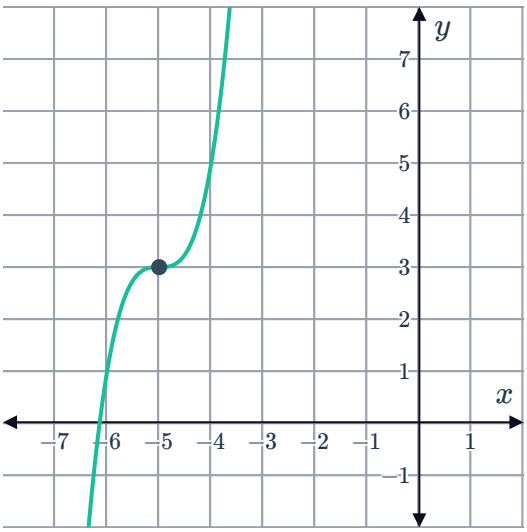
c $(-5, 3)$

d $(-5, 3)$ is a horizontal point of inflection.

e As $f''(x) = 0$ and the concavity changes from negative to positive, there is a point of inflection at $x = -5$. The point of inflection is $(-5, 3)$.

f As $x \to \infty, y \to \infty$ and as $x \to -\infty, y \to -\infty$.

g



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a $(0, 2)$

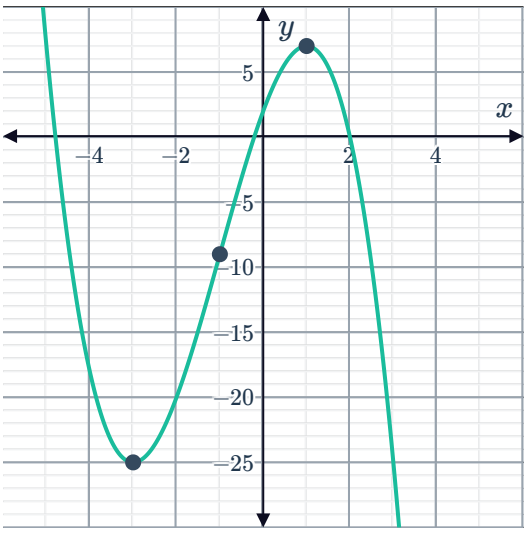
b $(-3, -25), (1, 7)$

c $(-3, -25)$ is a local minimum and $(1, 7)$ is a local maximum.

d As $f''(x) = 0$ and the concavity changes from positive to negative, there is a point of inflection at $x = -1$. The point of inflection is $(-1, -9)$.

e As $x \to \infty, y \to -\infty$ and as $x \to -\infty, y \to \infty$.

f



15

a $(0, -16)$

b $(-2, 0), (1, 0)$

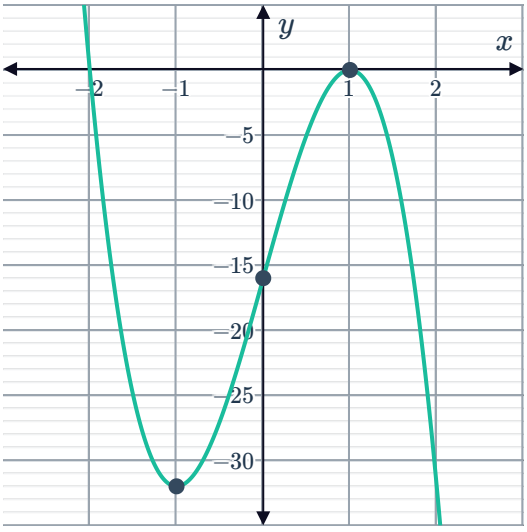
c $(-1, -32), (1, 0)$

d $(-1, -32)$ is a local minimum and $(1, 0)$ is a local maximum.

e As $f''(x) = 0$ and the concavity changes from positive to negative, there is a point of inflection at $x = 0$. The point of inflection is $(0, -16)$.

f As $x \to \infty, y \to -\infty$ and as $x \to -\infty, y \to \infty$.

g



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16 **a** $(0, 16)$

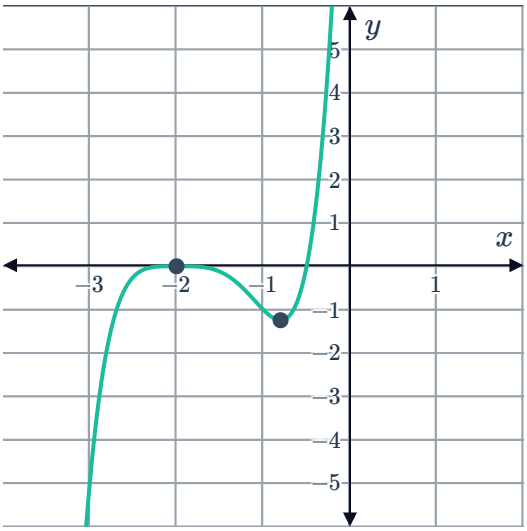
b $(-2, 0), \left(-\frac{1}{2}, 0\right)$

c $(-2, 0), \left(-\frac{4}{5}, -\frac{3888}{3125}\right)$

d $(-2, 0)$ is a local maximum and $\left(-\frac{4}{5}, -\frac{3888}{3125}\right)$ is a local minimum.

e As $x \to \infty, y \to \infty$ and as $x \to -\infty, y \to -\infty$.

f



17

a $(0, 2)$

b None

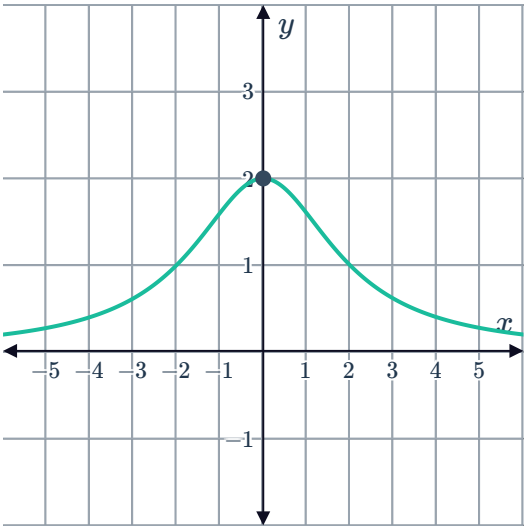
c Even

d $(0, 2)$

e $(0, 2)$ is the global maximum.

f As $x \to \infty, y \to 0$ and as $x \to -\infty, y \to 0$.

g



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